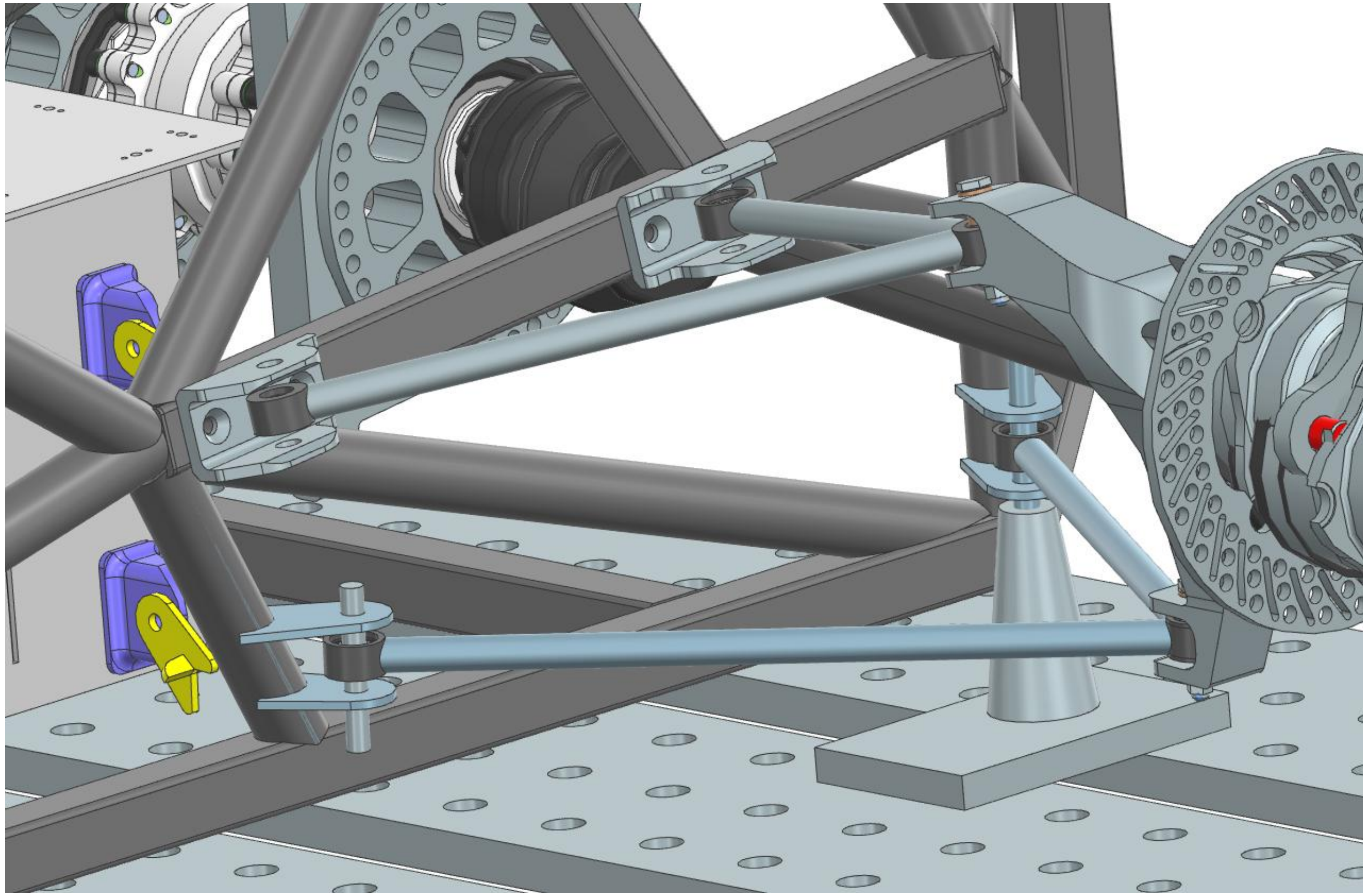
# A-arms Overview

Project: Design and machine A-arms, fixtures, and jigs

- **A-arm Tubes x16**
- **Weld Cups x28**
- **Inboard Brackets x3**
- **Tabs x8**
- **Tabs Jig x2**
- Gussets x8
- A-arm welding jig x1

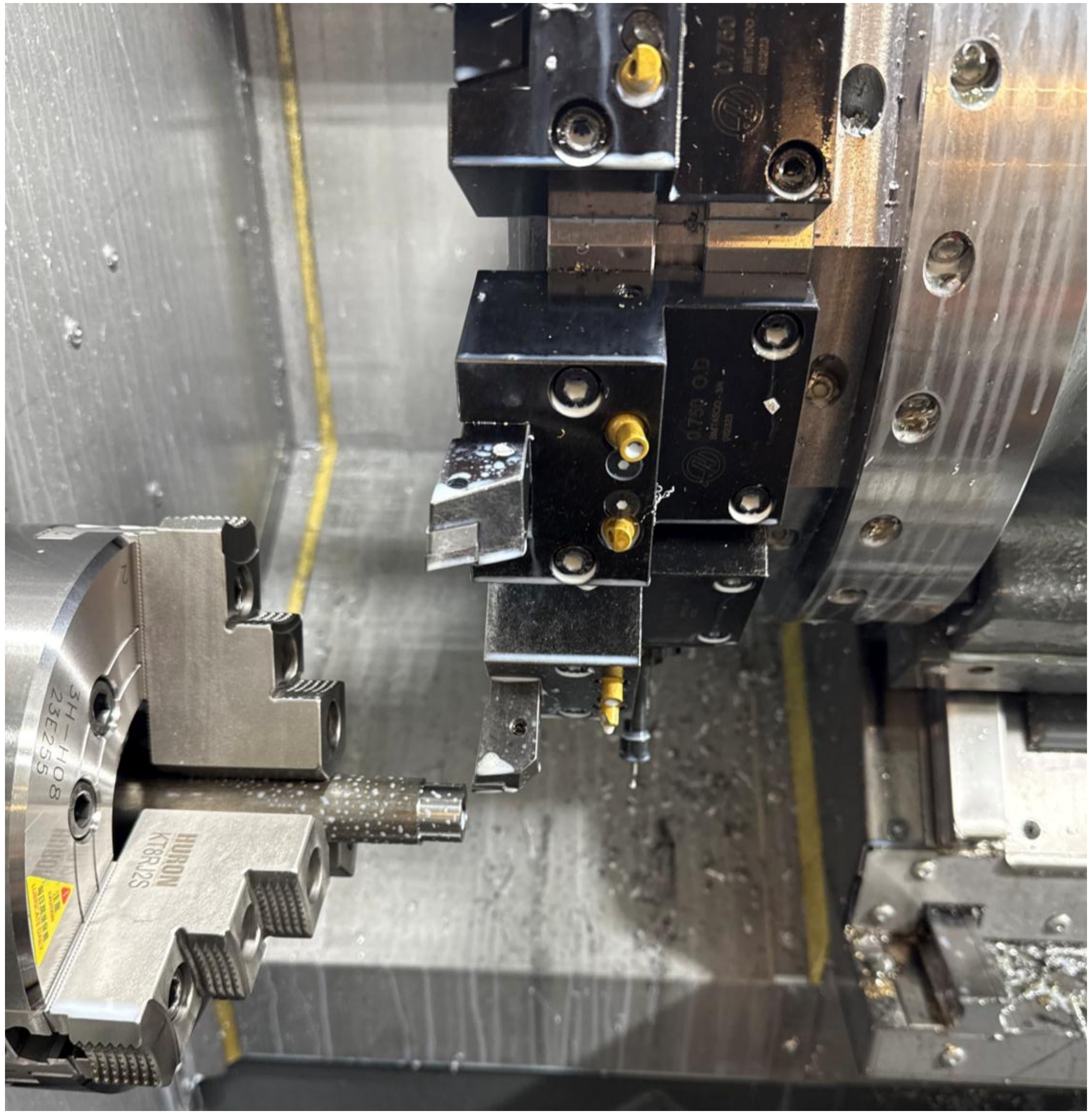


# A-arm Tubes

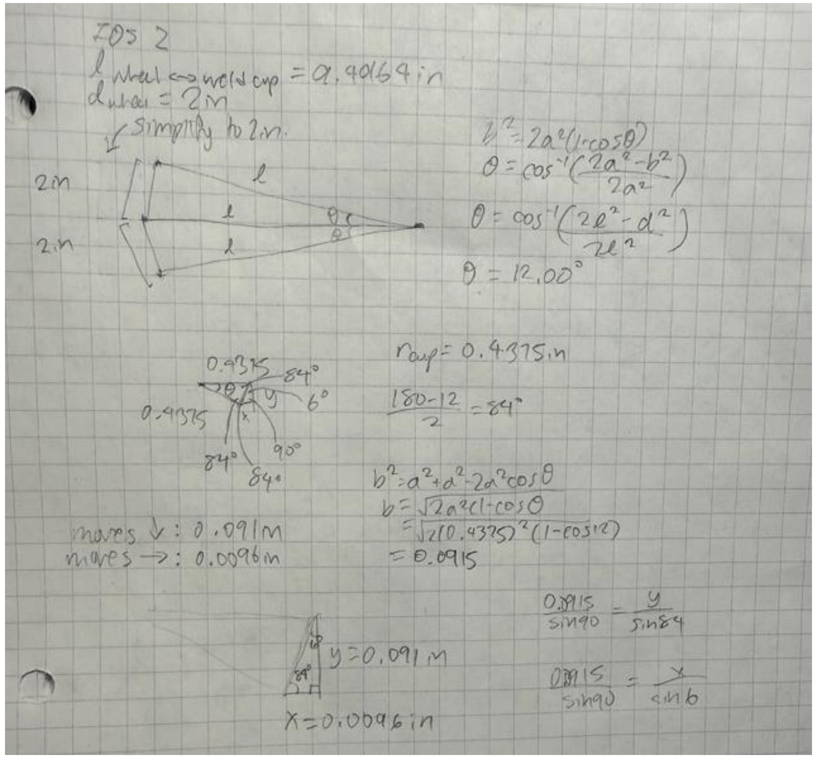
- Decreased machining time by ~30%
- Original method: support one end of the tube with clamp
- New method: remove need for clamping, simply move the tube by aligning it with the end mill
- Quantity: 16

# Weld Cups

- Haas CNC Lathe, bore gauge
- Tolerancing (.6560 to .6565)
- Manual lathe for deburring



| | A | B | C |
|---|---|---|---|
| 1 | Determining thickness of tubes | | |
| 2 | | | |
| 3 | Factor of Safety | 2 | |
| 4 | mg_car | 700 | lb |
| 5 | weight on one lower a-arm tube (mg_car)(.7)(1/2)(FOS) | 490 | lb |
| 6 | weight on upper a-arm (mg_car)(.3)(1/2)(FOS) | 210 | lb |
| 7 | | | |
| 8 | L_min | 5 | |
| 9 | L_max | 12 | |
| 10 | E (young's modulus) | 29,000,000 | psi |
| 11 | K | 1 | |
| 12 | | | |
| 13 | euler buckling formula: $P = (\pi^2 EI) / L^2$ | | |
| 14 | I=(P(KL)^2)/(Eπ^2) | | |
| 15 | minimum I for lower a-arm tube | 0.0002465 | |
| 16 | minimum I for upper a-arm tube | 0.0001057 | |
| 17 | | | |
| 18 | I=pi/64(OD^4-ID^4) | | |
| 19 | 0.50" OD × 0.035" wall | 0.00139 | |
| 20 | 0.75" OD × 0.065" wall | 0.00828 | |
| 21 | | | |

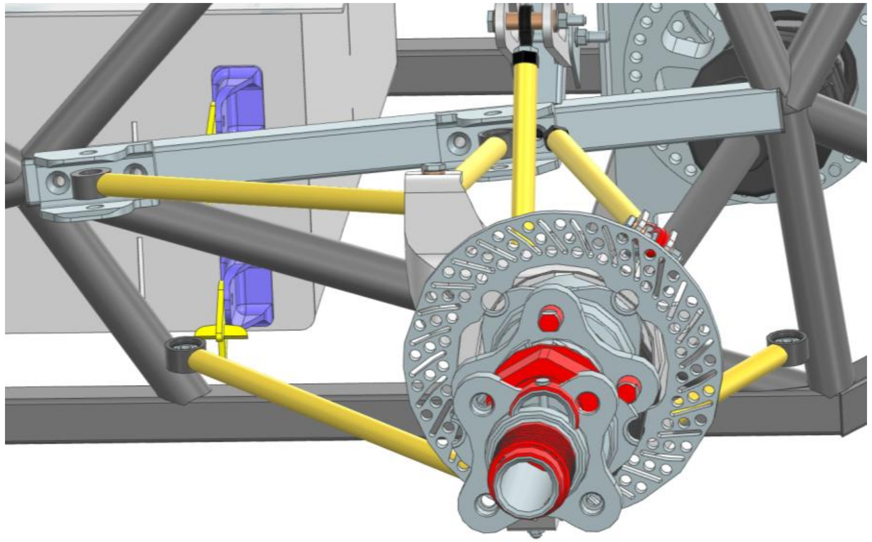



# Hand Calcs

- Buckling calculations
- A-arm hitting inboard brackets calculations

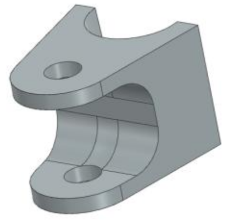
# Inboard Brackets

Had to create brackets to be welded on to steel tubes instead of bolted on like previous year's designs



# Inboard Brackets

Less sloped tube

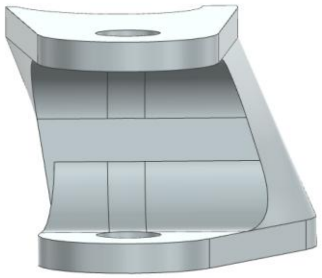


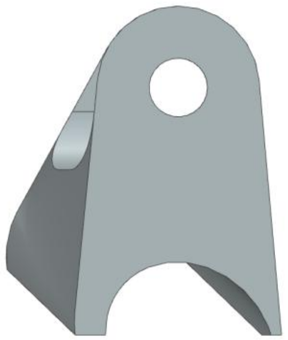
Challenge: Balancing weight and machinability

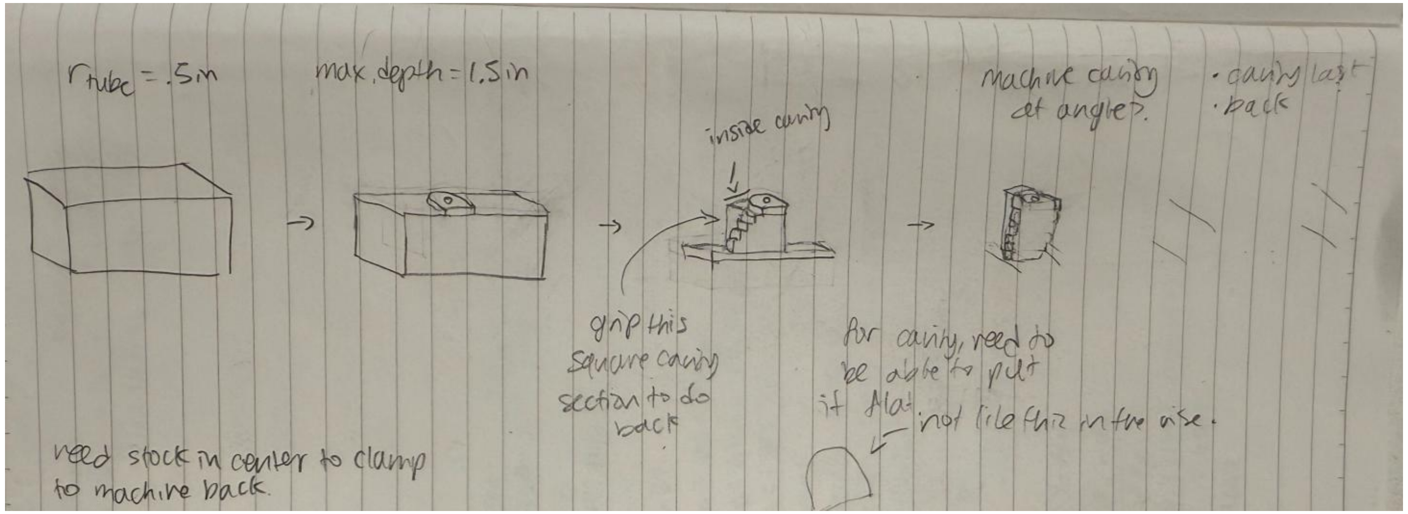
<u>More sloped tube</u>

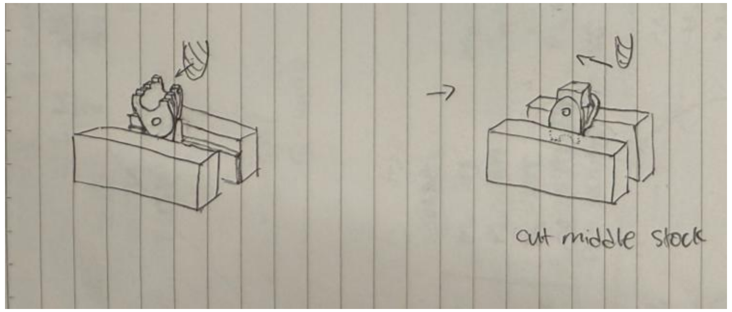
1) Bulky sides → make sloped 3D surface, but rough cut for faster machining since the look didn't matter
2) Tilted profile in the vise for cutting out cavity stock → have one of the steps be flat

Determining the order of these operations took some thought







# Tabs

- Made the mistake of assuming chief engineer's initial design was automatically the way to go; asked lead if doing tabs would be better but was told to just do brackets
- Assumed that welding tabs onto the tube would be unstable, but then asked chief engineer directly about tabs
- Realized I didn't know a lot about welding → learned how to weld, observed others doing different types of welds
- Lesson: don't just be focused on solving the problem in front of you, give a lot of thought into how valid the overall approach is

# Tabs

- Inboard bracket design (652 grams for four brackets) → tabs (249 grams for 8 tabs)
- After FEA, tabs decreased from .185” → .125” thickness, saving 94 grams
- Overall 76% decrease in weight, 1.1 pounds saved

**Calculations**

<u>Result max stress</u>: 6.0074 psi = 60.074 MPa

Yield stress of 4130 steel: 435 MPa

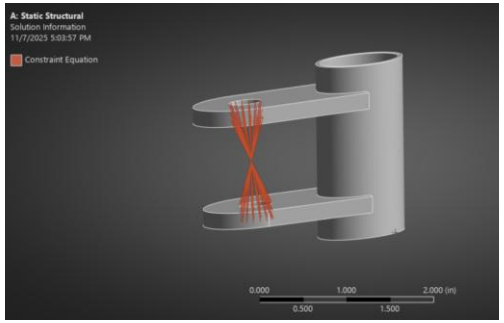
Factor of Safety
(435 Mpa)/1.6 = 271.875 MPa

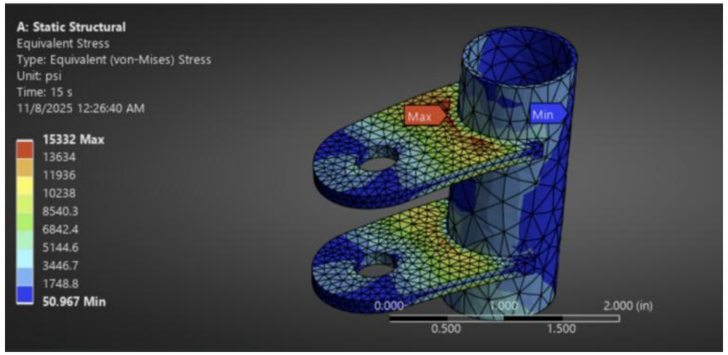
Weld strength
(271.875 MPa)(.8) = 217.5 MPa

Weld Penetration
(217.5 MPa)/2 = 108.75 MPa

<u>Must be under:</u> 108.75 MPa

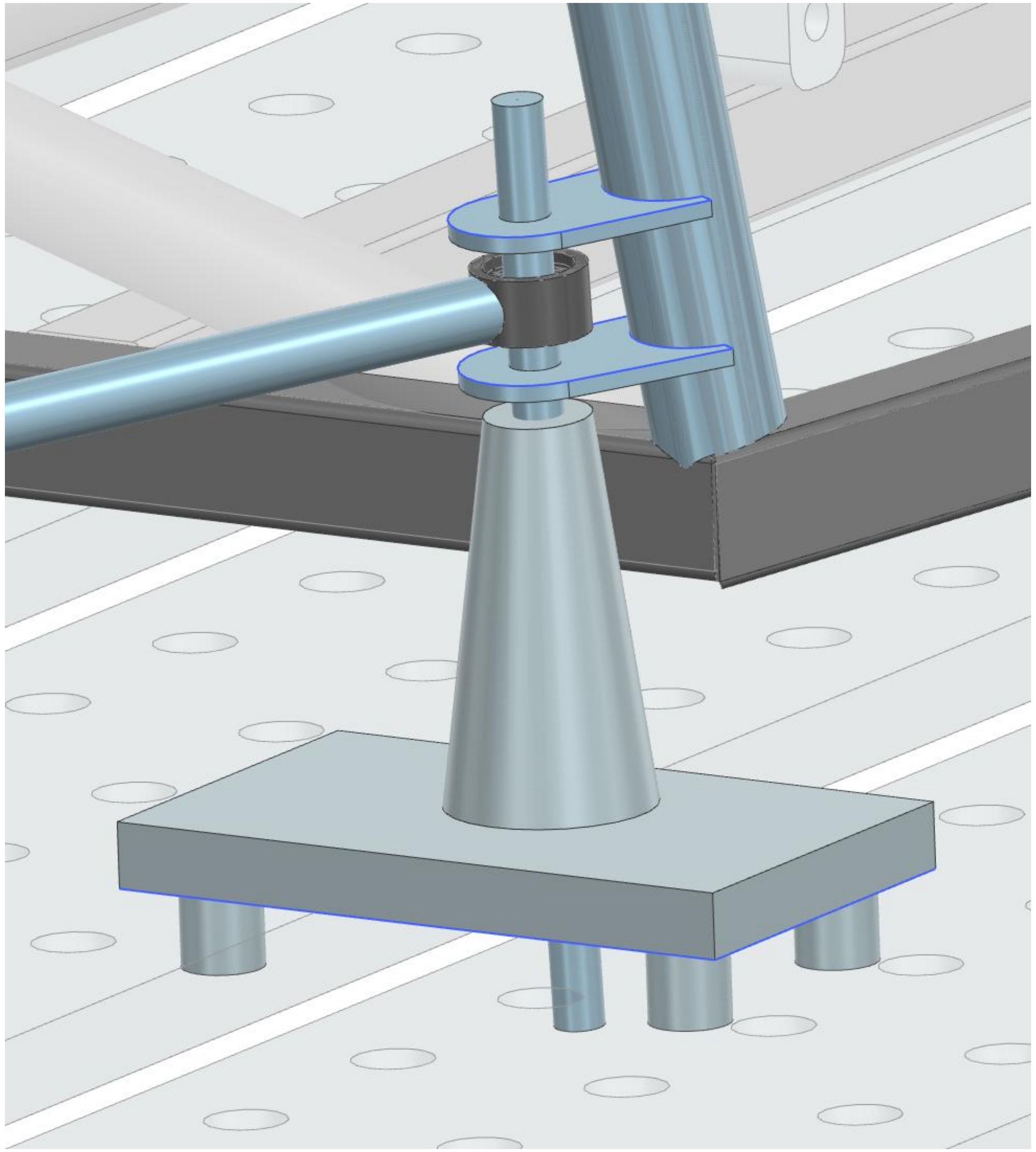
60.074 MPa < 108.75 MPa



# Tabs Jig

- 3D printed plastic jig
- Rod of same diameter as tab holes to keep it at the correct angle
- 3D print vertically to avoid the plastic deforming
- Nuts for fixing the tabs in place while welding and to distance steel from plastic to prevent burning
- 1” tube to correctly distance the tabs



# Lessons learned

- Knowledge in welding is also important for DFM, not just machining
- Make sure to simplify fully before trying to solve the problem
- I enjoy optimizing manufacturing processes and figuring out ways to machine a part